

Course Syllabus Part II
DSC 650 - Big Data
3 Credit Hours

Course Resources

Course Text(s)

Designing Data-Intensive Applications: The Big Ideas Behind Reliable, Scalable, and Maintainable Systems (1st Edition).

Martin Kleppmann

O'Reilly Media

ISBN-13: 978-1449373320

ISBN-10: 1449373321

Deep Learning with Python (1st Edition).

François Chollet

Manning Publications

ISBN-13: 978-1617294433

ISBN-10: 1617294438

Optional

Spark: The Definitive Guide: Big Data Processing Made Simple (1st Edition).

Bill Chambers & Matei Zaharia

O'Reilly Media

ISBN-13: 978-1491912218

ISBN-10: 1491912219

Required Resources

In this course, you will need to be able to:

- Access the Internet.
- Access Cyberactive
- Access to Github
- Collaborate Online via Video and Voice
- Collaborate while writing a single document
- Access to GitHub account
- Python programming environment using PyCharm, Anaconda, and Jupyter Notebook

Course Schedule

Week	Topic	Assigned Reading
1	Introduction to Big Data and Deep Learning	DDA - Chapter 1 DLP – Chapter 1

2	Data Models and Data Processing	<i>DDA</i> – Chapters 2 and 3 DLP – Chapter 2
3	Data Structures and Encoding	<i>DDA</i> – Chapters 2 and 3
4	Batch Processing Workflow	<i>DDA</i> – Chapter 10 DLP – Chapter 3
5	Machine Learning Fundamentals	DLP – Chapters 3 and 4
6	Computer Vision	DLP – Chapter 5
7	Replication and Partitioning	<i>DDA</i> – Chapters 5 and 6
8	Streaming, Messaging, and Transactions	<i>DDA</i> – Chapters 7, 9, and 11 (Optional) <i>DDA</i> – Chapter 8
9	Text and Sequence Processing	DLP – Chapter 6
10	The Future of Big Data	<i>DDA</i> – Chapter 12 DLP – Chapters 8 and 9

Course Activities

In this section of the syllabus, I will describe what we will be doing in each of the activities for each week. Specifically, I will be describing your deliverables – those items you need to submit at or before the deadline. You can find more detail on grading criteria for each category by viewing its detailed rubric.

Exercises

You will be assigned an exercise or series of exercises based on the weekly topic to complete and submit to the assignment link. These are not group assignments to complete and should be done on your own. However, if you have questions about a specific exercise, you are encouraged to use the discussion board to discuss with your classmates, without completing the assignment together.

Discussion/Participation

Every week you will be required to make 10 posts via an online platform. The goal is to simulate real world discussion and participation – there will not be formal posts required or required topics to discuss. There may be optional topics provided to start discussion, however, sharing information, troubleshooting, asking questions/feedback, etc. will be the primary focus for discussion/participation. Discussion/Participation will be graded as follows:

# of Posts	Percentage
0	0%
5	50%
10	100%

Grade and Point Breakdown

Component:	Percentage	Point Value	Number of Times	Total
Discussion/Participation	33.3%	50 Points	10	500
Exercises	66.6%	100 Points	10	1000
Total Points				1500

Late Work

Late work is not accepted unless arrangements are made with the instructor for very special, unavoidable circumstances. If you do not alert the professor before or shortly after something that will make you late, the chances of special arrangements are much lower. If in doubt, please email as soon as possible.

Participation

Students are required to login often and contribute to the class on a regular basis, including posting in the online platform, submitting assignments, and participating in group activities as required. If you have specific participation requirements related to your educational funding or student status, you are expected to monitor your own participation to ensure you are in compliance with those requirements.

Expectations for Students

- Students should expect to spend approximately 10-15 hours per week to complete the activities and assignments in this course.
- Students will log in as often as needed to complete their assignments and progress through the course.
- Students will treat their classmates and the instructor with respect and courtesy.
- Students are responsible for keeping current with the reading assignments and coming to class prepared to discuss the work assigned.
- Students are responsible for knowing what assignments are due and when.
- Students will submit only their own work and will not commit plagiarism or other acts of academic dishonesty.
- Students will contact the instructor as soon as personal problems arise that may affect the student's ability to complete assignments on time.

Expectations for Faculty

- The instructor will treat all students with respect and courtesy.
- The instructor will make grading criteria clear and follow the criteria scrupulously in evaluating student work.

- The instructor will provide feedback about student work within 6 days of due dates (or 24 hours prior to the next due date)—feedback that helps the student learn and improve.
- The instructor will respond to all student messages within 48 hours.